Lahore Traffic System Multimodal RAG

Retriever Evaluation Report

Methodology

Evaluation reuses the persisted LightRAG vector index and local Sentence Transformer embeddings. Ground-truth labels reference exact chunk IDs present in the persisted index.

Configuration

Dataset: 18 verified examples
Embedding: sentence-transformers/all-MiniLM-L6-v2 (384 dimensions)
Top-k: 10
LangSmith: Enabled (lahore-traffic-retriever-evaluation)

Aggregate Metrics

Metric	Value
precision@1	0.6111
precision@3	0.2963
precision@5	0.1889
precision@10	0.1000
recall@1	0.5833
recall@3	0.8333
recall@5	0.8889
recall@10	0.9444
mrr	0.7228
ndcg@3	0.7368
ndcg@5	0.7583
ndcg@10	0.7768
hit_rate	0.9444
empty_retrieval	0.0000

Limitations

The dataset currently contains 18 verified examples. Context precision and context recall are not claimed because answer-level context annotations are not yet part of the verified dataset. LangSmith traces and dataset are available under the project 'rag-retriever-evaluation'.